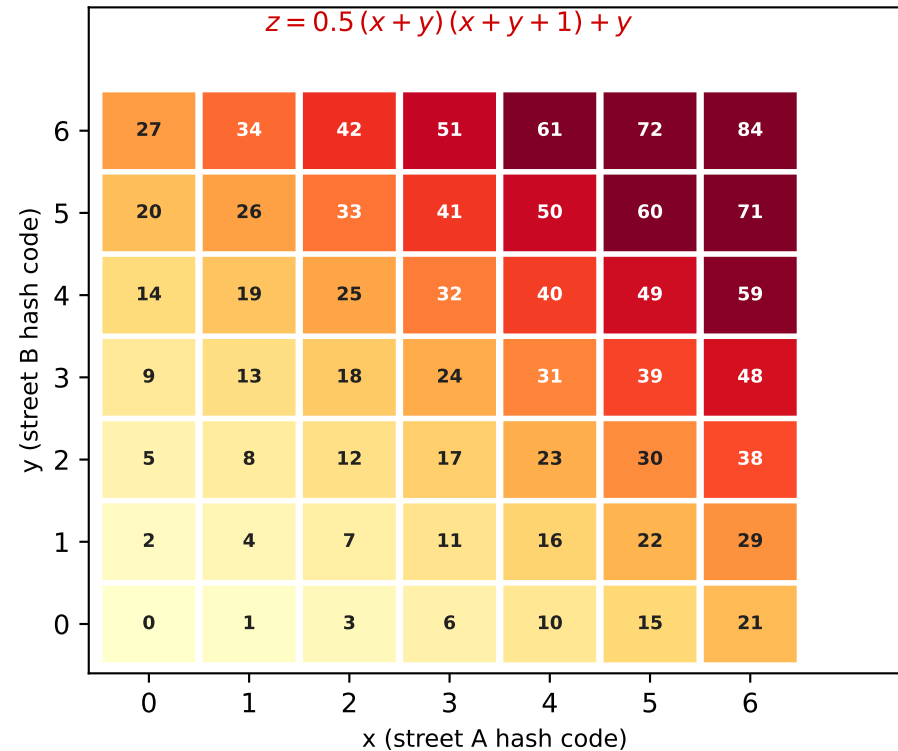


Cantor Pairing: unique key z for each (x, y) pair

$$z = 0.5(x + y)(x + y + 1) + y$$



**HashMap Cache Architecture
(avoids redundant routing API calls)**

